

A REAL TIME PROJECT REPORT
ON
WAILING SIREN USING 555 TIMER IC
Submitted in partial fulfillment of the requirements for the award of
BACHELOR OF TECHNOLOGY
IN
ELECTRONICS & COMMUNICATION ENGINEERING



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2024-2025

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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
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CERTIFICATE

This is to certify that this is a bonafide record of the project work entitled “**WAILING SIREN USING 555 TIMER IC**” is a bonafide work carried out by **V.YASHVANTH (23281A0480), A.MANIKANTA (23281A0474), R.VINAY (23281A0478), B.SAIKIRAN (23281A0483), R. NIKHIL (23281A0429)** in partial fulfillment of the requirements for the degree **BACHELOR OF TECHNOLOGY** in **ELECTRONICS & COMMUNICATION ENGINEERING** by the Jawaharlal Nehru Technological University,Hyderabad,during the academic year 2024-2025

Project Guide

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ACKNOWLEDGEMENT

The success of any course depends mostly on the teachers who teach us. Only good teaching can interpret the syllabus and produce desirable changes and competent citizens. This one was a team effort and many people whose names do not appear on the cover deserve credit. Primarily, we thank God almighty for his manifold mercies in carrying out of our project successfully.

We heart fully thank our principal **Dr. K. ESWARAIAH** for providing all the resources in completing our project.

We sincerely thank our Head of the Department of Electronics & Communication Engineering and Professor **Dr. B. Ramesh**, for encouraging us in doing real time projects and for his guidance.

We thank our project coordinator **Dr.M.Shyam sundar** Associate Professor in ECE for their continuous support in completion of this project.

We thank our internal guide **Mr.P.Anil Kumar**, Assistant Professors in department of Electronics and Communication Engineering, for his guidance and support.

We also thank other teaching and non-teaching staff of Kamala Institute of Technology and Science, for supporting us in every stage of our mini project, entitled “**WAILING SIREN USING 555 TIMER IC**”

We also thank our parents and friends for their moral support throughout the project work that helped to strengthen our will.

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ABSTRACT

This project aims to design, simulate, and construct a simple and cheap Wailing siren. This Wailing siren was implemented using 555 timer IC. The background information of the history and uses of Wailing siren, basic elements of Wailing siren and the theory behind how a basic wailing siren works are presented as well as the design criteria for the constructed Wailing siren. The simulation was done using workbench simulation software and the simulation results are presented in this thesis. The Wailing siren circuit was successfully constructed. This wailing siren was capable of evacuate emergency areas near to its region.

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CHAPTER-1

INTRODUCTION

1.1 BACKGROUND

In 1819, an improved siren was developed and named by Baron Charles Cagniard de la Tour. De la Tour's siren consisted of two perforated disks that were mounted coaxially at the outlet of a pneumatic tube. One disk was stationary, while the other disk rotated. The rotating disk periodically interrupted the flow of air through the fixed disk, producing a tone. De la Tour's siren could produce sound under water, suggesting a link with the sirens of Greek mythology; hence the name he gave to the instrument.

1.2 IMPORTANCE OF THE PROJECT

A siren is a loud noise-making device. Civil defense sirens are mounted in fixed locations and used to warn of natural disasters or attacks. Sirens are used on emergency service vehicles such as ambulances, police cars, and fire engines.

1.3 PROJECT AIM

The aim of the project is to design and construct simple and cheap wailing siren circuit. This wailing siren should be capable of producing sounds in emergency areas to evacuate.

1.4 TYPES OF WAILING SIREN

Sirens are used on emergency service vehicles such as ambulances, police cars, and fire engines. There are two general types: mechanical and electronic. Picking the right type based on budget and goals is the key to a successful journey in the sport of treasure hunting. Below is an overview of the different types of wailing siren.

1.4.1 MECHANICAL

A mechanical siren uses a rotor and stator to chop an air stream, which is forced through the siren by radial vanes in the spinning rotor. An example of this type of siren is the Federal Signal 2T22, which was originally developed during the Cold War and produced from the early 1950s to the late 1980s. This particular design employs dual rotors and stators to sound each pitch. Because the sound power output of this type of siren is the same in every

direction at all times, it is described as omnidirectional. The Federal 2T22 was also marketed in a 3-signal configuration known as the

FederalSignal3T22, with the capability for a "hi-lo" (High-Low) signal.

1.4.2 ELECTRONIC

Electronic sirens may be mechanically rotated to cover a wide area, or may have transducers facing in all directions to make an omnidirectional pattern. A directional siren may be applied where notification is only required for a defined area in one direction.

As with electromechanical sirens, there are both omi-directional, directional, and rotating categories, though Whelen Engineering produces sirens which oscillate through 360 degrees, rotating in one direction and then in the other to allow a hard-wired connection between the amplifiers and the siren drivers. These sirens can also be set to rotate any amount from 0 to 360 degrees, allowing sirens to broadcast.

CHAPTER-2

CONSTRUCTION AND WORKING OF WAILING SIREN

To design wailing siren we are using IC 555 timer, resistor, NPN transistor, capacitor and speaker. In this circuit, we are using the 555-timer IC in astable mode. Similar to most 555 timer circuits the IC pins 2 and 6 are connected and pin 4 is connected to the ground. A PNP transistor is connected as a switch between the supply voltage and pin 8 of the IC. The base of this transistor is connected to the 100uf capacitor through a 100K resistor. The output pin 3 of the IC is connected to the base of an NPN transistor. This transistor works as a switch for driving the 8-ohm output speaker.

2.1 WORKING

From the circuit, we can see that the IC is powered through a PNP transistor. Since it is a PNP transistor, it means the negative voltage at the gate will turn it on. Initially, when the supply is turned on, the capacitor starts to charge through the 100k and 220k resistor, and due to this fully charged capacitor, there is a positive voltage at the gate of the transistor which switches it off.

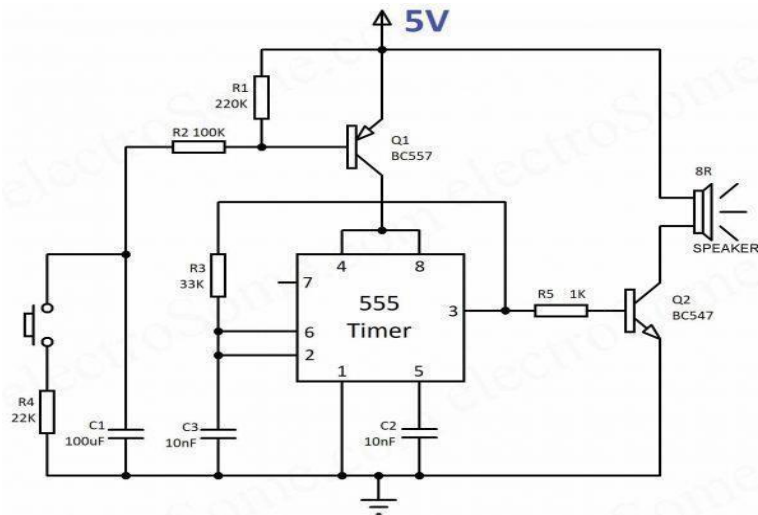


Fig:2.1 Wailing Siren Using 555 Timer Circuit Diagram

When we press the push button the capacitor finds a way to discharge through the 22k resistor which in result provides a negative voltage at the transistor that turns it on. This phenomenon creates a wailing siren because the charging and discharging of the capacitor takes time. This is the reason why the amplitude of the siren changes depending on the time of the button pressed.

2.2 PRINCIPLE

The main principle of this circuit is to produce a wailing siren. The 555 timer IC is operated in astable mode. When the switch is pressed, speaker produces high pitch siren and when it is released, its pitch decreases and is switched off after 30 seconds.

CHAPTER-3

ELEMENTS OF WAILING SIREN

3.1 IC 555 TIMER

3.1.1 INTRODUCTION OF IC 555 TIMER



Fig:3.1 IC 555 Timer

One of the most versatile linear IC's is the 555 timer which was first introduced in early 1970 by Stigmatic Corporation giving the name as SE/NE 555 timer. This IC is a monolithic timing circuit that can produce accurate and highly stable time delays or oscillation. Like other commonly used op-amps, this IC is also very much reliable easy to use and cheaper in cost. It has a variety of applications including monostable and astable multi vibrators, de-de converters, digital logic probes, waveform generators, analog frequency meters and tachometers, temperature measurement and control devices, voltage regulators etc. The timer basically operates in one of the two modes either as a monostable (one-shot) multi vibrator or as an astable (free-running multi vibrator. The SE 555 is designed for the operating temperature range from 55°C to 125 while the NE 555 operates over a temperature range of 0°C to 70 C.

3.1.2 THE IMPORTANT FEATURES OF THE 555 TIMER ARE:

It operates from a wide range of power supplies ranging from 5 Volts to + 18 Volts supply voltage

- Sinking or sourcing 200 mA of load current.
- The external components should be selected properly so that the timing intervals can be made into several minutes Proper selection of only a few external components allows timing intervals of several minutes along with the frequencies exceeding several hundred kilo hertz.
- It has a high current output: the output can drive TTL.
- It has a temperature stability of 50 parts per million (ppm) per degree Celsius change in temperature, or equivalently 0.005%/ °C.

- The duty cycle of the timer is adjustable with the maximum power dissipation per package is 600 mW and its trigger and reset inputs are logic compatible.

3.1.3 PIN CONFIGURATION



Fig:3.2 555 IC Timer Pin Configuration

The **555 Timer IC** is available as an 8-pin metal can, an 8-pin mini DIP (dual-in-package) Or a 14-pin DIP

We have seen that Multivibrators and CMOS Oscillators can be easily constructed from discrete components to produce relaxation oscillators for generating basic square wave output waveforms. But there are also dedicated IC's such as the 555 timer especially designed to accurately produce the required output waveform with the addition of just a few extra timing components.

One such device that has been around since the early days of IC's and has itself become something of an industry "standard" is the 555 Timer Oscillator which is more commonly called the "555 Timer".

The basic 555 timer gets its name from the fact that there are three internally connected 5kΩ resistors which it uses to generate the two comparators reference voltages. This IC consists of 23 transistors, 2 diodes and 16 resistors. The explanation of terminals out of the 555 timer IC is as follows. The pin number used in the following discussion refers to the 8-pin DIP and 8-pin metal can packages.

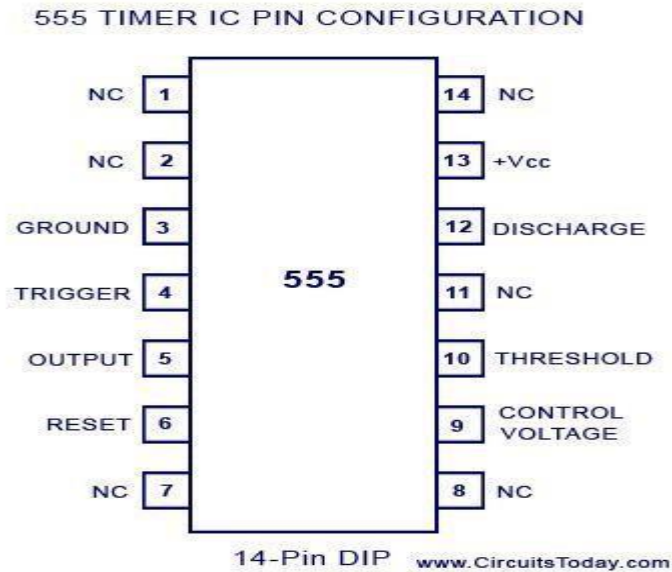


Fig 3.3 555 Timer IC Pin Configuration

Pin 1: Grounded Terminal: All the voltages are measured with respect to this terminal.

Pin 2: Trigger Terminal: This pin is an inverting input to a comparator that is responsible for transition of flipflop from set to reset. The output of the timer depends on the amplitude of the external trigger pulse applied to this pin.

Pin 3: Output Terminal: Output of the timer is available at this pin. There are two ways in which a load can be connected to the output terminal either between pin 3 and ground pin (pin 1) or between pin 3 and supply pin (pin 8). The load connected between pin 3 and ground supply pin is called the normally on load and that connected between pin 3 and ground pin is called the normally off load.

Pin 4: Reset Terminal: To disable or reset the timer a negative pulse is applied to this pin due to which it is referred to as reset terminal. When this pin is not to be used for reset purpose, it should be connected to + VCC to avoid any possibility of false triggering. .

Pin 5: Control Voltage Terminal: The function of this terminal is to control the threshold and trigger levels. Thus either the external voltage or a pot connected to this pin determines the pulse width of the output waveform. The external voltage applied to this

pin can also be used to modulate the output waveform. When this pin is not used, it should be connected to ground through a 0.01 micro Farad to avoid any noise problem.

Pin 6: Threshold Terminal: This is the non-inverting input terminal of comparator 1. which compares the voltage applied to the terminal with a reference voltage of $\frac{2}{3} V_{CC}$. The amplitude of voltage applied to this terminal is responsible for the set state of flipflop.

Pin 7: Discharge Terminal: This pin is connected internally to the collector of transistor and mostly a capacitor is connected between this terminal and ground. It is called discharge terminal because when transistor saturates, capacitor discharges through the transistor. When the transistor is cut-off, the capacitor charges at a rate determined by the external resistor and capacitor.

Pin 8: Supply Terminal: A supply voltage of + 5 V to + 18 V is applied to this terminal with respect to ground(pin 1).

3.1.4 MODES

The 555 timer can operate in 3 different modes:

1. Monostable Mode
2. Astable Mode or free running
3. Bistable Mode or Schmitt Trigger

3.1.5 555 TIMER - MONOSTABLE MODE

Figure shows a Monostable 555 Timer circuit. The Monostable circuit outputs one pulse for each high to low transition of the trigger pin. The output will return to the "stable" state after a period of time that is a function of the attached resistor and capacitor values.

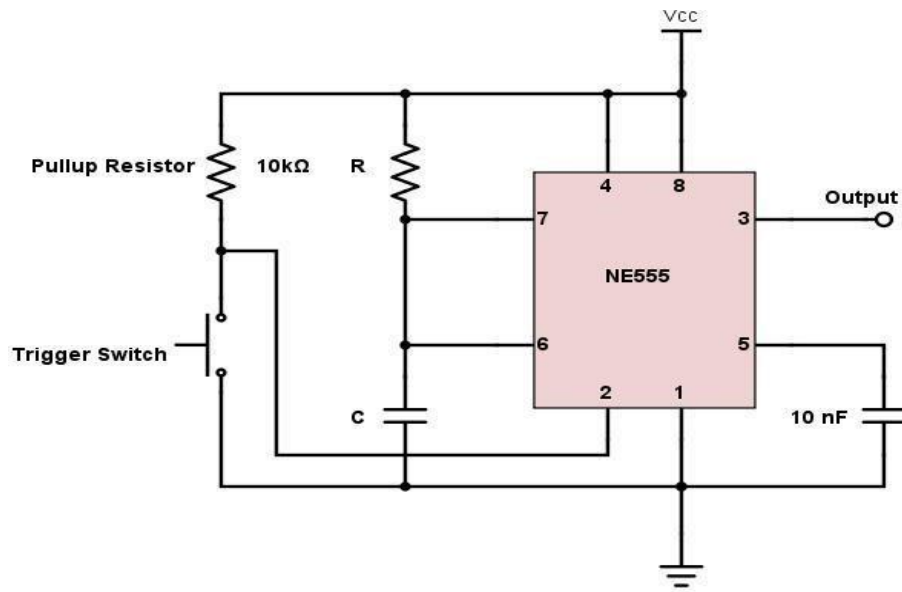


Fig:3.4 555 Timer Monostable Configuration

The discharge pin is normally connected to ground (internally). When the trigger pin transitions from V_{cc} to $33\% V_{cc}$ the discharge pin is then internally disconnected from ground and the output pin is set high. The capacitor C then starts to charge through resistor R . The threshold pin detects when the voltage across the capacitor reaches $66.6\% V_{cc}$. When the voltage across the capacitor reaches $66.6\% V_{cc}$, the output pin is set low and the discharge pin is connected back to ground. Finally, when the discharge pin is connected back to ground, the capacitor is discharged and the default "stable" state is reached.

The length of the output pulse depends on when the capacitor reaches $66.6\% V_{cc}$. This rate is determined by the charge capacity of the capacitor, C , and resistance, R . The length of the output pulse, t_P , is: $t_P = 1.1RC$. In short, starting with a discharged capacitor, the output pin is set high once the trigger pin transitions from high to low, and it remains high (the output pin) until the capacitor reaches 66% of V_{cc} .

The monostable 555 timer circuit can be used in the following applications:

- a) Debounce a momentary/pushbutton switch
- b) Turning on an actuator for a set period of time
- c) Turn an output from a resistive sensor from analog signal to digital signal. (Example: A potentiometer is a variable resistor. A project requires determining the position of a

potentiometer used for user input. All analog to digital (A/D) converters on a microcontroller have already been used but there are some digital inputs and outputs available. The potentiometer can be used as the resistor in a monostable 555 timer circuit. The microcontroller can then trigger the monostable circuit and measure t_P (since t_P now depends On the position of the potentiometer).

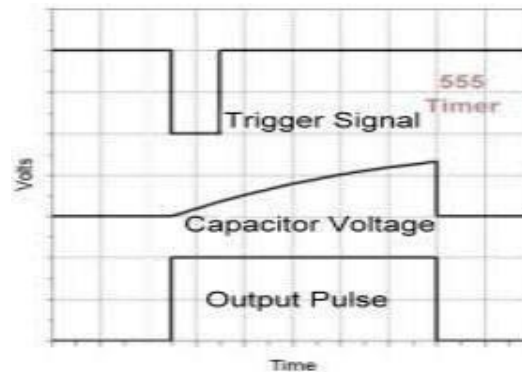


Fig:3.5 555 IC Timer-Astable Mode

3.1.6 ASTABLE MODE:

An astable 555 timer circuit the astable 555 timer circuit outputs a series of Pulses, alternating between two states. When the circuit is first turned on, the discharge pin is disconnected from ground (internally) and the output pin is set high since the trigger pin is below 33.3% V_{cc} (no charge in the capacitor). The capacitor, C , starts to charge through resistors R_1 and R_2 . The threshold pin detects when the voltage across the capacitor reaches 66.6% V_{cc} . When the voltage across the capacitor reaches 66.6% V_{cc} , the output pin is set low and the discharge pin is connected back to ground. When the discharge pin is connected back to ground, the capacitor starts discharging through resistor R_2 until the capacitor reaches 33.3% V_{cc} . The cycle then repeats and creates a multi vibrator is a form of the oscillator which has two states due to which we can get output by either of the states. It is basically two amplifier circuits arranged with regenerative feedback. In this, neither of the transistors conducts simultaneously. At a time, only one transistor conducts and another one is in off state. Some circuits have certain states; the state which has fast transition is called switching processes, where there is fast change of current as well as voltage. This switching is termed as triggering. Hence, we can trigger the circuit internally or externally.

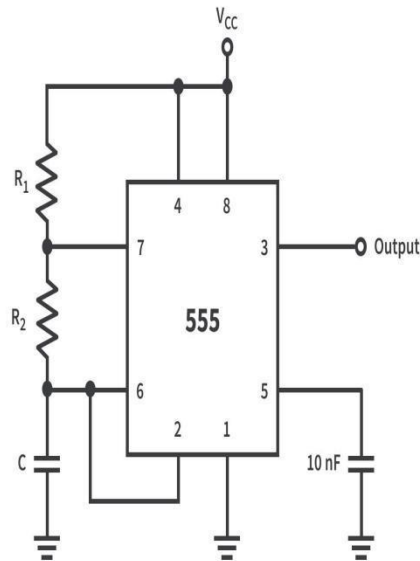


Fig:3.6 555 IC Timer Astable Configuration

Compared to the Monostable circuit, the Astable circuit connects the trigger pin to the capacitor and adds a resistor between the discharge and threshold pins. And Astable circuit triggers from previous output pulse whereas a monostable circuit requires an externally applied trigger

In short, the output pin oscillates from high to low creating a series of pulses as the capacitor charge oscillates from 33.3% to 66.6% V_{cc} , without any external triggers. The duration the output pin stays high, t_{HIGH} , is given below: $t_{HIGH} = 0.693 C (R_1 + R_2)$

The duration the output pin stays low, t_{LOW} is given below

$$t_{LOW} = 0.693 C R_2$$

The frequency, f , of the series of pulses is: $f = 1 / (t_{HIGH} + t_{LOW})$. The Astable 555 timer circuit can be used in the following applications:

- a) Modulate transmitters such as ultrasonic and IR transmitters
- b) Create an accurate clock signal (Example: There is a pulse accumulator pin on the 68HC11 microcontroller that counts pulses. You can apply an Astable 555 timer circuit set at 1 Hz frequency to the pulse accumulator pin and create a seconds counter within the microcontroller. The pulse accumulator will be covered in later in the course).
- c) Turn on and off an actuator at set time intervals for a fixed duration.

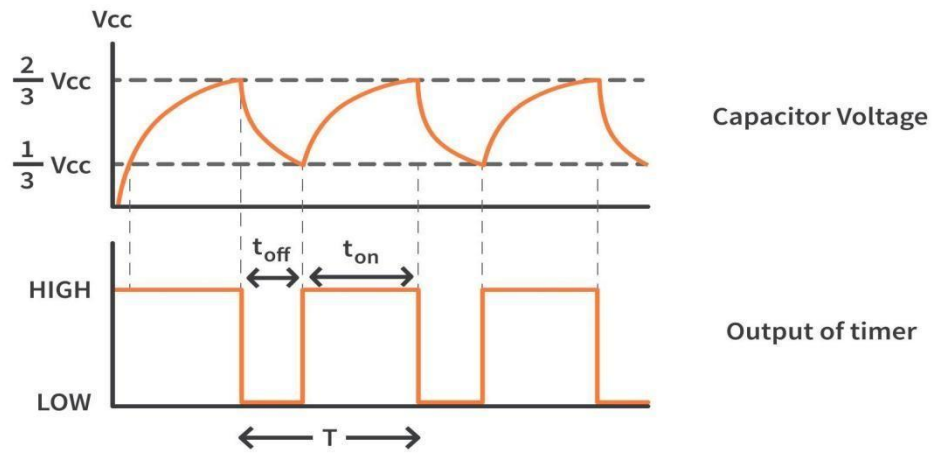


Fig:3.7 Waveform InAstable Mode

3.1.7 BISTABLE MODE:

In bistable mode, the 555 timer act as a basic flipflop. The trigger and reset inputs (pins 2 and 4 respectively on a 555 are held high via pull-up resistors while the threshold input(pin 6) is grounded. Thus configured, pulling the trigger momentarily to ground acts as a set and transitions the output in (pin 3) to V_{cc} (high state). Pulling the reset input to ground acts as a reset and transition the output pin to ground(low state). No timing capacitors are required in a bistable configuration. Pin 5 (discharge) is left unconnected, or may be used as an open collector output.

A 555 timer can be used to create a Schmitt trigger which converts a noisy input into a clean digital output. The input signal should be connected through a series capacitor which then connects to the trigger and threshold pins. A resistor divider, from V_{cc} to GND, is connected to the previous tied pins. The reset pin is tied to V_{cc} .

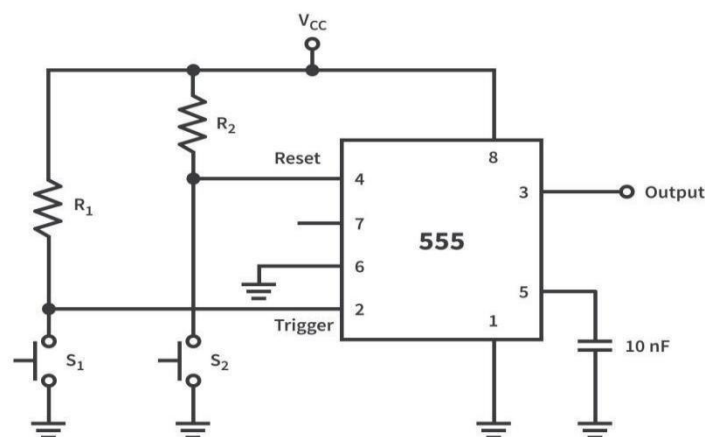


Fig 3.8 Schematic ofa 555 In Bistable Mode

3.1.8 SPECIFICATIONS

These specifications apply to the original bipolar NE555. Other 555 timers can have different specifications depending on the grade (industrial, military, medical, etc.).

Table 3.1

Part number	NE555
IC Process	Bipolar
Supply voltage (V_{CC})	4.5 to 16 V
Supply current ($V_{CC} = +5\text{ V}$)	3 to 6 mA
Supply current ($V_{CC} = +15\text{ V}$)	10 to 15 mA
Output current (maximum)	200 mA
Maximum Power dissipation	600 mW
Power consumption (minimum operating)	30 mW @ 5 V, 225 mW @ 15 V
Operating temperature	0 to 70 °C

3.1.9 Block diagram

Represented with a block diagram it consists of 2 comparators, a flip-flop, a voltage divider, a discharge transistor and an output stage.

The voltage divider consists of three identical 5k resistors which create two reference voltages at 1/3 and 2/3 of the supplied voltage, which can range from 5 to 15V.

Next are the two comparators. A comparator is a circuit element that compares two analog input voltages at its positive (non-inverting) and negative (inverting) input terminal. If the input voltage at the positive terminal is higher than the input voltage at the negative terminal the comparator will output 1. Vice versa, if the voltage at the negative input terminal is higher than the voltage at the positive terminal, the comparator will output 0. The first comparator negative input terminal is connected to the $2/3$ reference voltage at the voltage divider and the external “control” pin, while the positive input terminal to the external “Threshold” pin

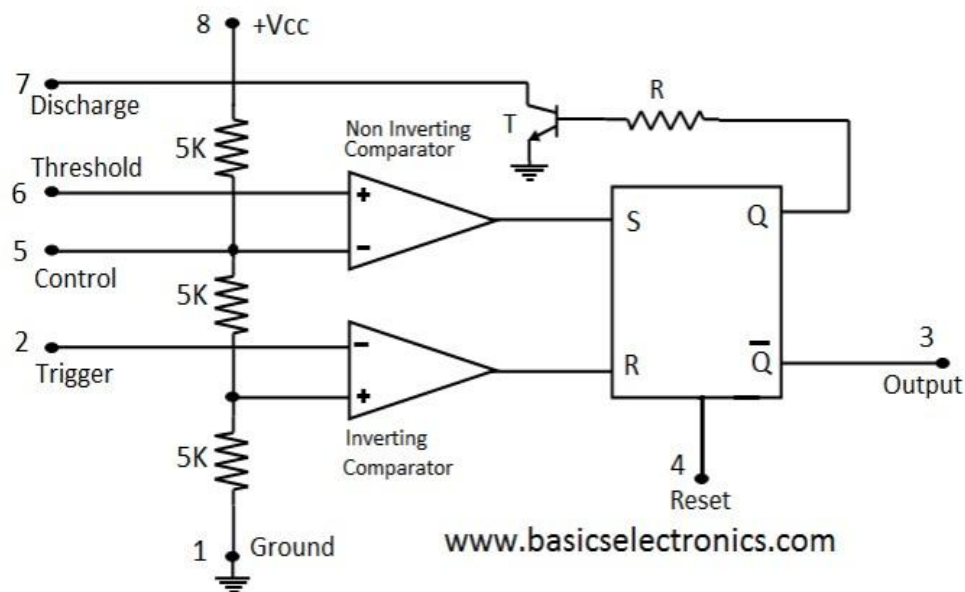


Fig.:3.9 Block Diagram of IC 555

On the other hand, the second comparator negative input terminal is connected to the “Trigger” pin, while the positive input terminal to the $1/3$ reference voltage at the voltage divider.

So using the three pins, Trigger, Threshold and Control, we can control the output of the two comparators which are then fed to the R and S inputs of the flip-flop. The flipflop will output 1 when R is 0 and S is 1, and vice versa, it will output 0 when R is 1 and S is 0. Additionally the flip-flop can be reset via the external pin called “Reset” which can override the two inputs, thus reset the entire timer at any time.

The Q-bar output of the flip-flop goes to the output stage or the output drivers which can either source or sink a current of 200mA to the load. The output of the flip-flop is also connected to a transistor that connects the “Discharge” pin to ground.

3.2 RESISTOR

A resistor is a passive two-terminal electrical component that implements electrical resistance as a circuit element. In electronic circuits, resistors are used to reduce current flow, adjust signal levels, to divide voltages, bias active elements, and terminate transmission lines, among other uses. High-power resistors that can dissipate many watts of electrical power as heat may be used as part of motor controls, in power distribution systems, or as test loads for generators. Fixed resistors have resistances that only change slightly with temperature, time or operating voltage. Variable resistors can be used to adjust circuit elements (such as a volume control or a lamp dimmer), or as sensing devices for heat, light, humidity, force, or chemical activity.

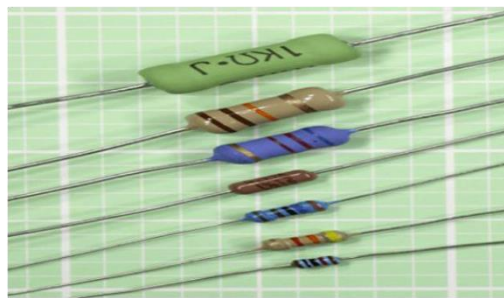


Fig:3.10 Resistors

Resistors are common elements of electrical networks and electronic circuits and are ubiquitous in electronic equipment. Practical resistors as discrete components can be composed of various compounds and forms. Resistors are also implemented within integrated circuits.

The electrical function of a resistor is specified by its resistance: common commercial resistors are manufactured over a range of more than nine orders of magnitude. The nominal value of the resistance falls within the manufacturing tolerance, indicated on the component.

Ohm's law

Ohm's law is one of the most popular and important laws that help us define the relationship between voltage, current and resistance. The law was first established by a

German physicist named Georg Simon Ohm and was the most important theory that described the quantitative characteristics of the physics of electricity.

Ohm's law can be also be considered as an empirical law. It can be used in developing conclusions or in reasoning while conducting many experiments especially in showing that current for some materials is approximately proportional to the electric field. We shall learn more about this law including its definition, formulas, applications and more in this lesson.

$$I=V/R$$

Carbon composition resistors (CCR) consist of a solid cylindrical resistive element with embedded wire leads or metal end caps to which the lead wires are attached. The body of the resistor is protected with paint or plastic. Early 20th-century carbon composition resistors had uninsulated bodies; the lead wires were wrapped around the ends of the resistance element rod and soldered. The completed resistor was painted for color coding of its value.

3.2.1 Carbon resistors:

The resistive element in carbon composition resistors is made from a mixture of finely powdered carbon and an insulating material, usually ceramic. A resin holds the mixture together. The resistance is determined by the ratio of the fill material (the powdered ceramic) to the carbon. Higher concentrations of carbon, which is a good conductor, result in lower resistances. Carbon composition resistors were commonly used in the 1960s and earlier, but are not popular for general use now as other types have better specifications, such as tolerance, voltage dependence, and stress. Carbon composition resistors change value when stressed with over-voltages. Moreover, if internal moisture content, such as from exposure for some length of time to a humid environment, is significant, soldering heat creates a non-reversible change in resistance value. Carbon composition resistors have poor stability with time and were consequently factory sorted to, at best, only 5% tolerance. These resistors are non-inductive, which provides benefits when used in voltage pulse reduction and surge protection applications. Carbon composition resistors have higher capability to withstand overload relative to the component's size.

Carbon composition resistors are still available, but relatively expensive. Values ranged from fractions of an ohm to 22 Mega ohms. Due to their high price, these resistors

are no longer used in most applications. However, they are used in power supplies and welding controls. They are also in demand for repair of vintage electronic equipment where authenticity is a factor.

3.3 CAPACITOR

A capacitor is a passive two-terminal electrical component that stores potential energy in an electric field. The effect of a capacitor is known as capacitance. While some capacitance exists between any two electrical conductors in proximity in a circuit a capacitor is a component designed to add capacitance to a circuit. The capacitor was originally known as a condenser or compensator. The original name still widely used in many language, but not in English

A capacitor consists of two conductors separated by a non-conductive region. The non-conductive region can either be a vacuum or an electrical insulator material known as a dielectric. Examples of dielectric media are glass, air, paper, plastic, ceramic, and even a semiconductor depletion region chemically identical to the conductors. From Coulomb's law, a charge on one conductor will exert a force on the charge carriers within the other conductor, attracting opposite polarity charge and repelling like polarity charges, thus an opposite polarity charge will be induced on the surface of the other conductor. The conductors thus hold equal and opposite charges on their facing surfaces and the dielectric develops an electric field.



Fig:3.11 Capacitors

3.3.1 Feature of capacitors

I. Capacitor Characteristics – Nominal Capacitance, (C)

The nominal value of the **Capacitance**, C of a capacitor is the most important of all capacitor characteristics. This value measured in pico-Farads (pF), nano-Farads (nF) or micro-Farads (μ F) and is marked onto the body of the capacitor as numbers, letters or coloured bands.

The capacitance of a capacitor can change value with the circuit frequency (Hz) y with the ambient temperature. Smaller ceramic capacitors can have a nominal value as low as one pico-Farad, (1pF) while larger electrolytic's can have a nominal capacitance value of up to one Farad, (1F).

All capacitors have a tolerance rating that can range from -20% to as high as +80% for aluminium electrolytic's affecting its actual or real value. The choice of capacitance is determined by the circuit configuration but the value read on the side of a capacitor may not necessarily be its actual value.

ii. Capacitor Characteristics – Working Voltage, (WV)

The **Working Voltage** is another important capacitor characteristic that defines the maximum continuous voltage either DC or AC that can be applied to the capacitor without failure during its working life. Generally, the working voltage printed onto the side of a capacitors body refers to its DC working voltage, (WVDC).

DC and AC voltage values are usually not the same for a capacitor as the AC voltage value refers to the r.m.s. value and NOT the maximum or peak value which is 1.414 times greater. Also, the specified DC working voltage is valid within a certain temperature range, normally -30°C to +70°C.

Any DC voltage in excess of its working voltage or an excessive AC ripple current may cause failure. It follows therefore, that a capacitor will have a longer working life if operated in a cool environment and within its rated voltage. Common working DC voltages are 10V, 16V, 25V, 35V, 50V, 63V, 100V, 160V, 250V, 400V and 1000V and are printed onto the body of the capacitor.

iii. Capacitor Characteristics – Tolerance, ($\pm\%$)

As with resistors, capacitors also have a **Tolerance** rating expressed as a plus-minus value either in picofarad's ($\pm\text{pF}$) for low value capacitors generally less than 100pF or as a percentage ($\pm\%$) for higher value capacitors generally higher than 100pF.

The tolerance value is the extent to which the actual capacitance is allowed to vary from its nominal value and can range anywhere from -20% to +80%. Thus a 100 μF capacitor with a $\pm 20\%$ tolerance could legitimately vary from 80 μF to 120 μF and still remain within tolerance.

Capacitors are rated according to how near to their actual values they are compared to the rated nominal capacitance with coloured bands or letters used to indicate their actual tolerance. The most common tolerance variation for capacitors is 5% or 10% but some plastic capacitors are rated as low as $\pm 1\%$.

iv. Capacitor Characteristics – Leakage Current

The dielectric used inside the capacitor to separate the conductive plates is not a perfect insulator resulting in a very small current flowing or “leaking” through the dielectric due to the influence of the powerful electric fields built up by the charge on the plates when applied to a constant supply voltage.

This small DC current flow in the region of nano-amperes (nA) is called the capacitors **Leakage Current**. Leakage current is a result of electrons physically making their way through the dielectric medium, around its edges or across its leads and which will over time fully discharge the capacitor if the supply voltage is removed.

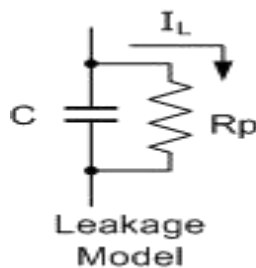


Fig:3.12 Leakage of Capacitor

When the leakage is very low such as in film or foil type capacitors it is generally referred to as “insulation resistance” (R_p) and can be expressed as a high value resistance in parallel with the capacitor as shown. When the leakage current is high as in

electrolytic's it is referred to as a "leakage current" as electrons flow directly through the electrolyte.

Capacitor leakage current is an important parameter in amplifier coupling circuits or in power supply circuits, with the best choices for coupling and/or storage applications being Teflon and the other plastic capacitor types (polypropylene, polystyrene, etc) because the lower the dielectric constant, the higher the insulation resistance.

Electrolytic-type capacitors (tantalum and aluminium) on the other hand may have very high capacitances, but they also have very high leakage currents (typically of the order of about 5-20 μA per μF) due to their poor isolation resistance, and are therefore not suited for storage or coupling applications. Also, the flow of leakage current for aluminium electrolytic's increases with temperature.

V .Capacitor Characteristics – Working Temperature, (T)

Changes in temperature around the capacitor affect the value of the capacitance because of changes in the dielectric properties. If the air or surrounding temperature becomes too hot or too cold the capacitance value of the capacitor may change so much as to affect the correct operation of the circuit. The normal working range for most capacitors is -30°C to $+125^{\circ}\text{C}$ with nominal voltage ratings given for a **Working Temperature** of no more than $+70^{\circ}\text{C}$ especially for the plastic capacitor types.

Generally for electrolytic capacitors and especially aluminium electrolytic capacitor, at high temperatures (over $+85^{\circ}\text{C}$ the liquids within the electrolyte can be lost to evaporation, and the body of the capacitor (especially the small sizes) may become deformed due to the internal pressure and leak outright. Also, electrolytic capacitors can not be used at low temperatures, below about -10°C , as the electrolyte jelly freezes.

Vi .Temperature Coefficient, (TC)

The **Temperature Coefficient** of a capacitor is the maximum change in its capacitance over a specified temperature range. The temperature coefficient of a capacitor is generally expressed linearly as parts per million per degree centigrade (PPM/ $^{\circ}\text{C}$), or as a percent change over a particular range of temperatures. Some capacitors are non linear (Class 2 capacitors) and increase their value as the temperature rises giving them a temperature coefficient that is expressed as a positive "P".

Some capacitors decrease their value as the temperature rises giving them a temperature coefficient that is expressed as a negative “N”. For example “P100” is +100 ppm/°C or “N200”, which is -200 ppm/°C etc. However, some capacitors do not change their value and remain constant over a certain temperature range, such capacitors have a zero temperature coefficient or “NPO”. These types of capacitors such as Mica or Polyester are generally referred to as Class 1 capacitors.

Most capacitors, especially electrolytic’s lose their capacitance when they get hot but temperature compensating capacitors are available in the range of at least P1000 through to N5000 (+1000 ppm/°C through to -5000 ppm/°C). It is also possible to connect a capacitor with a positive temperature coefficient in series or parallel with a capacitor having a negative temperature coefficient the net result being that the two opposite effects will cancel each other out over a certain range of temperatures. Another useful application of temperature coefficient capacitors is to use them to cancel out the effect of temperature on other components within a circuit, such as inductors or resistors etc.

Polarization

Capacitor **Polarization** generally refers to the electrolytic type capacitors but mainly the Aluminium Electrolytic’s, with regards to their electrical connection. The majority of electrolytic capacitors are polarized types, that is the voltage connected i.e. positive to positive and negative to negative. Incorrect polarization can cause the oxide layer inside the capacitor to break down resulting



Fig:3.13 Polarization of Capacitor

in very large currents flowing through the device resulting in destruction as we have mentioned earlier.

The majority of electrolytic capacitors have their negative, -ve terminal clearly marked with either a black stripe, band, arrows or chevrons down one side of their body as shown, to prevent any incorrect connection to the DC supply. Some larger electrolytic's have their metal can or body connected to the negative terminal but high voltage types have their metal can insulated with the electrodes being brought out to separate spade or screw terminals for safety.

Also, when using aluminium electrolytic's in power supply smoothing circuits care should be taken to prevent the sum of the peak DC voltage and AC ripple voltage from becoming a "reverse voltage".

3.4 SPEAKER

Speaker gives the sound when the metal is placed near the coil. 8 ohms speaker is used as wailing siren.

A loudspeaker (or "speaker") is an electro acoustic transducer that produces sound in response to an electrical audio signal input. Non-electrical loudspeakers were developed as accessories to telephone systems, but electronic amplification by vacuum tube made loudspeakers more generally useful. The most common form of loudspeaker uses a paper cone supporting a voice coil electromagnet acting on a permanent magnet, but many other types exist. Where accurate reproduction of sound is required, multiple loudspeakers may be used, each reproducing a part of the audible frequency range. Miniature loudspeakers are found in devices such as radio and TV receivers, and many forms of music players. Larger loudspeaker systems are used for music, sound reinforcement in theatres and concerts, and in public address system.



Fig:3.14 Speaker

Terminology, The term "loudspeaker" may refer to individual transducers (known as "drivers") or to complete speaker systems, consisting of an enclosure adding one or more drivers.

To adequately reproduce a wide range of frequencies most loudspeaker systems employ more than one driver, particularly for higher sound pressure level or maximum accuracy. Individual drivers are used to reproduce different frequency ranges. The drivers are named subwoofers (for very low frequencies); woofers (low frequencies), mid-range speakers (middle frequencies); tweeters (high frequencies) and sometimes super tweeters, optimized for the highest audible frequencies.

The terms for different speaker drivers differ, depending on the application. In way systems there is no mid-range driver, so the task of reproducing the mid-range sounds falls upon the woofer and tweeter. Home stereos use the designation Tweeter for the high frequency driver, while professional concert systems may designate them as "HF" or "highs". When multiple drivers are used in a system, a "crossover network" called a crossover, separates the incoming signal into different frequency ranges and routes them to the appropriate driver. A loudspeaker system with separate frequency bands is described as "n-way speakers". A two-way system will have a woofer and a tweeter, a three-way system employs a woofer, a mid-range and a tweeter. Electrical characteristics of a dynamic loudspeaker.

A speaker introduction involves establishing the person's credibility, motivating audience interest, and saying what the speaker could not say. Not many speakers will jump to the stage and share their list of accomplishments, as this would appear arrogant and could quickly turn off an audience. At the same time, if you are able to share that they have turned two companies around and would like to share lessons learned, your audience may see the value in giving their attention. Being designated to introduce a speaker is an honor and an important duty that requires planning and preparation.

Scot Ober states, "Remarks should be directed at welcoming the speaker and establishing his or her qualifications to speak on the topic" (Ober, S., 1995). You may start with a quote from their work, or a quote from a publication or colleague describing

them. You may decide to use humor. All these options are available, but whatever you choose, let respect and dignity be your overriding goal. The function and role of the introduction is to focus the spotlight squarely on the speaker. You should not distract the audience from that task with your dress, gestures, antics, or by talking about yourself.

The person you are introducing may already be well known to the audience, but you can always find some new information to share. You may need to consider the unusual, or the little known, when introducing someone who is famous. You may also consider mentioning their most recent work or activity as it relates to the topic of the presentation.

Avoid the “laundry list” approach to a summary of their education and experience, as this may bore the audience. Instead, focus on something specific and relevant. Your range of options is almost limitless, but your time frame and overall function are not. You need to be brief, and you need to establish the speaker’s credibility while motivating interest.

According to Bonnie Devet, “Performing the role of introducer also reinforces the rhetorical principles seminal to any business writing course: the need for ethos (credibility of both speakers and introducers), for audience-based discourse, and for accuracy” (Devet, B., 1995). Think of an introduction as a speech in miniature. Your purpose is to inform, your time frame is (typically) one to three minutes, and your specific purpose is to inform the audience about the speaker’s qualifications, credibility, and enthusiasm for the topic he or she will cover.

3.5 POWER SUPPLY

The 9V battery is an extremely common battery that was first used in transistor radios. It features a rectangular prism shape that utilizes a pair of snap connectors which



Fig: 3.15 Battery

A wide array of both large and small battery manufacturers produce versions of the 9V battery. Possible chemistries of primary (non-rechargeable) 9V batteries include Alkaline, Carbon-Zinc (Heavy Duty), Lithium. Possible chemistries of secondary (rechargeable) 9V batteries include nickel-cadmium (NiCd), nickel-metal hydride (NiMH), and lithium ion. The performance and application of the battery can vary greatly between different chemistries, meaning that some chemistries are better suited for some applications over others.

CHAPTER-4

ADVANTAGES & APPLICATIONS

4.1 ADVANTAGES

- The loud noise created by the siren can make it easy for the emergency responders to spot your house in no time.
- Siren provides alertness from danger, it gives enough time to the occupants to escape instead of individually going to each person to ask them to escape.
- Siren systems make them suitable as mass warning and notification systems for various types of hazards and disasters.
- They are easy to monitor and to control remotely and they offer a high level of reliability through relying on various energy sources.

4.2 APPLICATIONS

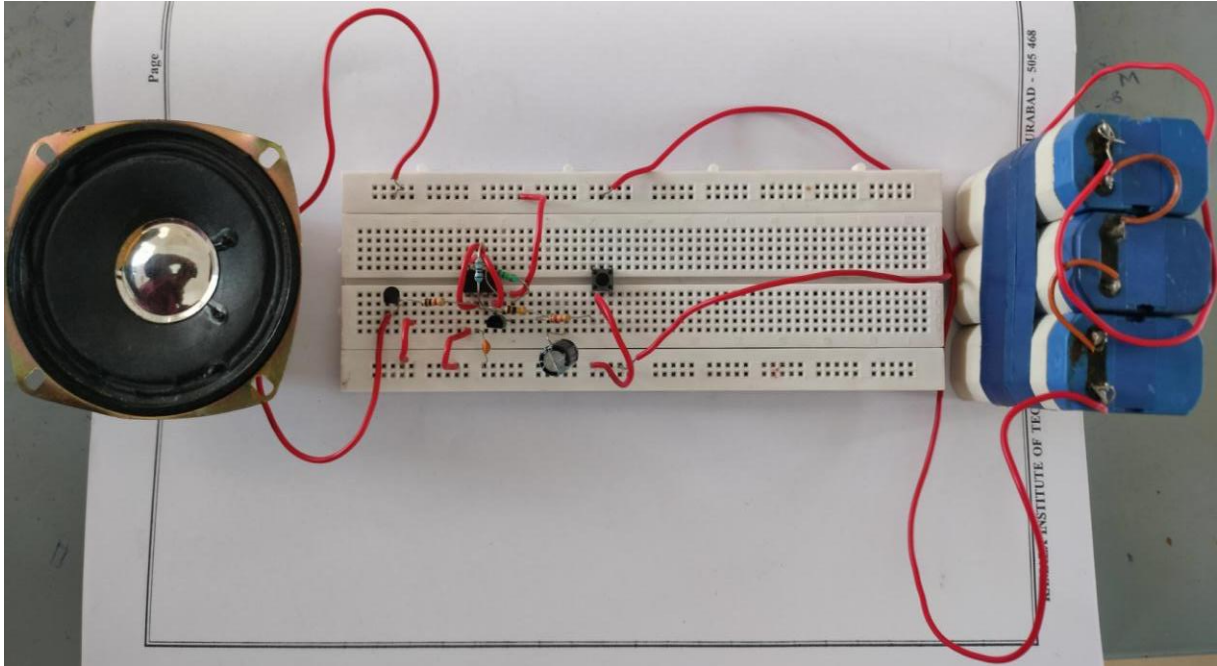
- This circuit with slight modifications can be used at the time of disaster to warn. This can also be used in security applications.
- This can also be used in alarm circuits.
- This can also be used in toys.
- This circuit can be used as an alarm in disaster management system.
- It can be used as a signal for completion for some task.
- It can be used in offices for medication lunch breaks. It can be used at the airports to drive away the birds from the runway.

4.3 LIMITATIONS

This cannot be used in applications where the sound is required to switch off instantaneously.

CHAPTER-5

RESULT



CHAPTER-6

CONCLUSION

Hence we can conclude that wailing siren circuit is very useful because of its quick response and it is simple to design and it has many applications like security purpose and emergency purpose etc. And with slight modifications we can use in various applications.

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